PINNs for Options Pricing

FYS5429 – Advanced Machine Learning and Data Analysis for the Physical Sciences

STATUS REPORT

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STATUS REPORT AND PROGRESS

This section summarizes the current state of the project and outlines the planned work for the remaining time before the submission deadline of June 1, 2026.

Current Status (March 2026)

The theoretical foundation of the project is in place. The introduction covers the Black–Scholes PDE and its connection to the heat equation, the Heston stochastic volatility model, and the motivation for physics-informed approaches. The methods section establishes the PINN loss formulation, the MSE–likelihood connection, and the rationale for activation function selection. No numerical experiments have been run yet; the results and conclusion sections are pending implementation.

Milestones

Phase 1 — Black–Scholes PINN (March 21 – April 4)

Implement a PINN solver for the one-dimensional Black–Scholes PDE in PyTorch. Train on collocation points in the (S, t) domain and enforce the terminal payoff and boundary conditions through the composite loss in eq. (6). Validate option prices and first-order Greeks (Δ, Θ) against the analytical Black–Scholes formula.

Phase 2 — Activation Function Study (April 4 – April 18)

Systematically compare smooth activation functions (`tanh`, Swish, GELU, Softplus, SIREN/Sine) on the Black–Scholes problem. Evaluate convergence speed, final loss, and accuracy of second-order Greeks (Γ, Vomma). Produce summary tables and convergence plots.

Phase 3 — Heston PINN (April 18 – May 9)

Extend the PINN framework to the two-dimensional Heston PDE over (S, v, t). Address the additional challenges of the mixed derivative term $\partial^2 V/\partial S\,\partial v$ and the Feller boundary condition on the variance process. Compare results to a reference finite-difference or Monte Carlo solution.

Phase 4 — Calibration (May 9 – May 23)

Formulate the inverse problem: infer the Heston parameters $(\kappa, \theta, \xi, \rho, v_0)$ from a set of observed option prices by minimizing a data-fit term added to the PINN loss. Experiment with both synthetic and real market data.

Phase 5 — Write-up and Submission (May 23 – June 1)

Complete the results section with figures and tables. Write the conclusion, connecting findings back to the research questions. Final proof-read, bibliography check, and submission.

Key Risks

Training stability is the main practical concern: PINN losses for financial PDEs can be poorly conditioned due to the wide range of option values across the domain. Loss weighting and adaptive sampling strategies may be needed. The Heston calibration phase is the most experimental part of the project and may be descoped or simplified if the forward solver proves insufficiently accurate.

Abstract

This project investigates the use of Physics-Informed Neural Networks (PINNs) for solving and calibrating option pricing models in quantitative finance. We begin with the Black–Scholes partial differential equation (PDE), demonstrating how its equivalence to the heat equation motivates a PINN-based solver that embeds the governing differential operator directly into the training loss. The neural network is trained to satisfy both the PDE residual and the terminal payoff condition, and its prices and Greeks are compared against the known analytical solution. We then extend the framework to the Heston stochastic volatility model, which introduces a higher-dimensional PDE with a mixed derivative term and no simple closed-form solution. The PINN approach is shown to scale naturally to this setting. We further explore the use of PINNs for the inverse calibration problem: inferring Heston model parameters from observed option prices. The results illustrate the promise of physics-informed machine learning as a flexible, mesh-free solver for financial PDEs, while also exposing practical challenges related to training stability and loss balancing in multi-dimensional settings.

1 Introduction

The classical starting point for many partial differential equation (PDE) methods in mathematical finance is the heat equation, which describes the diffusion of temperature in a homogeneous medium. In one spatial dimension, the heat equation is given by

$$\frac{\partial u}{\partial t}(x,t) = \alpha \frac{\partial^2 u}{\partial x^2}(x,t),$$

where $u(x,t)$ represents temperature, t denotes time, x is the spatial coordinate, and $\alpha > 0$ is the thermal diffusivity constant. The equation states that the rate of change in temperature is proportional to the spatial curvature of the temperature field, capturing the fundamental mechanism of diffusion.

The importance of the heat equation in option pricing arises from the fact that the Black–Scholes PDE can be transformed into a diffusion equation of this form through a suitable change of variables.

Under the standard geometric Brownian motion model for an asset price S_t,

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

the value $V(S,t)$ of a European option satisfies the Black–Scholes PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0.$$

By introducing the log-price transformation

$$x = \ln\left(\frac{S}{K}\right), \quad \tau = T - t,$$

together with an exponential rescaling of the dependent variable, the Black–Scholes equation can be reduced to the standard diffusion equation

$$\frac{\partial u}{\partial \tau} = \frac{\partial^2 u}{\partial x^2}.$$

This connection provides both analytical insight and a natural bridge to modern numerical approaches. In this thesis, Physics-Informed Neural Networks (PINNs) are used to solve the Black–Scholes PDE by embedding the governing differential equation directly into the loss function, allowing neural networks to approximate option prices while respecting the underlying diffusion structure inherited from the heat equation.

While the Black–Scholes model assumes constant volatility, empirical financial markets exhibit stochastic volatility dynamics. A widely used extension is the Heston stochastic volatility model, in which both the asset price and its variance evolve as stochastic processes:

$$dS_t = rS_t dt + \sqrt{v_t}\, S_t dW_t^S,$$

$$dv_t = \kappa(\theta - v_t)dt + \xi\sqrt{v_t}\, dW_t^v,$$

with correlation

$$dW_t^S dW_t^v = \rho dt.$$

In this setting, the option price becomes a function $V(S,v,t)$ and satisfies the two-dimensional Heston PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2}vS^2\frac{\partial^2 V}{\partial S^2} + \rho\xi vS\frac{\partial^2 V}{\partial S \partial v} + \frac{1}{2}\xi^2 v\frac{\partial^2 V}{\partial v^2}$$
$$+ \quad rS\frac{\partial V}{\partial S} + \kappa(\theta - v)\frac{\partial V}{\partial v} - rV = 0.$$

Compared to the Black–Scholes equation, the Heston PDE introduces an additional spatial dimension and a mixed derivative term, making classical numerical solution methods more computationally demanding. This increased complexity makes the model particularly well-suited for physics-informed machine learning approaches. In this thesis, PINNs are therefore applied not only to the Black–Scholes equation but also to the Heston PDE, demonstrating how neural-network-based solvers can handle higher-dimensional option pricing problems governed by diffusion-type equations.

1.1 The Black–Scholes Model

The Black–Scholes model (Abdelmessih 2026) prices a European call or put option analytically under the assumption of constant volatility. Given the log-moneyness

variables d_1 and d_2,

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T},$$

the call and put prices are

$$C = SN(d_1) - Ke^{-rT}N(d_2),$$

$$P = Ke^{-rT}N(-d_2) - SN(-d_1),$$

where S is the current asset price, K the strike, r the risk-free rate, σ the volatility, T the time to maturity, and $N(\cdot)$ the standard normal CDF.

The Greeks

The sensitivity of the option price to its inputs — the Greeks — are partial derivatives of V with respect to market and model parameters. The first- and second-order Greeks are summarized in table 1, together with selected third-order Greeks. The Black–Scholes PDE can be written compactly in terms of the first- and second-order Greeks as

$$\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma + rS\Delta - rV = 0,$$

which makes explicit that the time decay Θ is balanced by gamma and delta exposure.

Table 1. Selected Black–Scholes Greeks.

Greek	Expression	Order
Δ	$\partial V/\partial S$	1st
Θ	$\partial V/\partial t$	1st
ρ	$\partial V/\partial r$	1st
Φ	$\partial V/\partial q$	1st
Vega	$\partial V/\partial \sigma$	1st
Γ	$\partial^2 V/\partial S^2$	2nd
Vomma	$\partial^2 V/\partial \sigma^2$	2nd
Vanna	$\partial^2 V/\partial S\,\partial \sigma$	2nd
Charm	$\partial^2 V/\partial S\,\partial t$	2nd
Speed	$\partial^3 V/\partial S^3$	3rd
Zomma	$\partial^3 V/\partial S^2\,\partial \sigma$	3rd
Color	$\partial^3 V/\partial S^2\,\partial t$	3rd

2 Methods

This section develops the methodological framework for the project. We first review the connection between maximum likelihood estimation and mean squared error as the natural training objective for neural networks. We then describe the Physics-Informed Neural Network (PINN) architecture and its application to the Black–Scholes and Heston PDEs.

Modern quantitative finance sits at the intersection of stochastic modeling, numerical methods, and machine learning. Classical stochastic volatility modeling builds on multiscale diffusion techniques, where volatility evolves on multiple time scales (Fouque et al. 2011). Analytical approaches to option pricing under stochastic dynamics are further developed in tractable model frameworks (Gulisashvili 2012), while probabilistic interpretations of derivative pricing link option values to risk-neutral expectations (Prefetta n.d.). Foundational deep learning theory and architectures are presented in Goodfellow et al. (2016), while practical implementations using modern software frameworks are discussed in Raschka et al. (2022). Together, these perspectives motivate physics-informed machine learning methods for solving option pricing problems governed by PDEs.

2.1 *From Maximum Likelihood to Mean Squared Error*

$$P(y \mid x; w) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\hat{y})^2}{2\sigma^2}\right) \tag{1}$$

$$L(w) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i-\hat{y}_i^2}{2\sigma^2}\right) \tag{2}$$

$$\ln L(w) = \sum_{i=1}^{n} \left[\ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{(y_i-\hat{y}_i)^2}{2\sigma^2}\right] \tag{3}$$

$$\text{Maximize:} \sum_{i=i}^{n} -\frac{(y_i-\hat{y}_i)^2}{2\sigma^2} \tag{4}$$

$$\text{Minimize:} \sum_{i=1}^{n} (y_i-\hat{y}_i)^2 \tag{5}$$

Assuming Gaussian noise on the targets and maximizing the log-likelihood is therefore equivalent to minimizing the mean squared error. This justifies MSE as the standard regression loss for neural networks, and it is the starting point for the composite PINN loss described below.

2.2 *Physics-Informed Neural Networks*

A Physics-Informed Neural Network (PINN) is a neural network $\hat{V}_\theta(S,t)$ trained to satisfy both observed data and the residual of a governing PDE. Rather than discretizing the domain on a mesh, the PDE is enforced at a set of collocation points sampled from the interior and boundaries of the domain. The total loss is a weighted sum of three terms:

$$\mathcal{L}(\theta) = \lambda_{\text{pde}}\,\mathcal{L}_{\text{pde}} + \lambda_{\text{bc}}\,\mathcal{L}_{\text{bc}} + \lambda_{\text{ic}}\,\mathcal{L}_{\text{ic}}, \tag{6}$$

where

$$\mathcal{L}_{\text{pde}} = \frac{1}{N_r}\sum_{i=1}^{N_r}\left|\mathcal{F}[\hat{V}_\theta](S_i,t_i)\right|^2, \tag{7}$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_b}\sum_{i=1}^{N_b}\left|\hat{V}_\theta(S_i^b,t_i^b) - V_b\right|^2, \tag{8}$$

$$\mathcal{L}_{\text{ic}} = \frac{1}{N_0}\sum_{i=1}^{N_0}\left|\hat{V}_\theta(S_i,T) - \max(S_i-K,0)\right|^2. \tag{9}$$

Here $\mathcal{F}[\hat{V}_\theta]$ denotes the PDE operator applied to the network output (the residual), V_b encodes the boundary conditions (e.g., $V(0,t) = 0$ and $V(S,t) \to S$ as $S \to \infty$ for a call), and the initial condition enforces the terminal payoff at $t = T$. The partial derivatives required to evaluate $\mathcal{F}$ are computed via automatic differentiation (Goodfellow et al. 2016).

For the Black–Scholes PDE the residual is

$$\mathcal{F}[\hat{V}] = \frac{\partial \hat{V}}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 \hat{V}}{\partial S^2} + rS\frac{\partial \hat{V}}{\partial S} - r\hat{V}.$$

For the Heston PDE the residual extends to two spatial dimensions (S, v), introducing an additional variance dimension and a mixed derivative term.

2.3 Activation Functions

The choice of activation function affects both the expressivity of the network and the quality of the automatic-differentiation graph used to compute PDE residuals. Standard choices such as ReLU are not suitable for PINNs because they yield zero second derivatives almost everywhere. The hyperbolic tangent $\tanh(z)$ is a natural choice: it is smooth, bounded, and its derivatives are analytically well-behaved, which is important when the loss includes second-order terms such as $\Gamma = \partial^2 V/\partial S^2$. Other smooth activations — Swish, GELU, Softplus, Sine (SIREN) — will also be explored and compared.

3 Results

These are the results

4 Conclusion

References

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5 Appendix